Implement Client Script & UI Policy (Incident)

Project Description:

The "Implement Client Script & UI Policy (Incident)" project focuses on automating business logic and enhancing user experience within ServiceNow IT Service Management (ITSM). By leveraging JavaScript-based Client Scripts and configurable UI Policies, the system dynamically controls form behavior, field visibility, mandatory requirements, and read-only states on Incident forms based on user roles and input values in real-time.

This project streamlines incident logging and resolution workflows by eliminating manual validation errors and enforcing standard operating procedures directly at the UI layer. It ensures that critical data such as priority, assignment group, and resolution notes are accurately captured before submission, thereby optimizing service desk efficiency and compliance.

Scenarios

Scenario 1: A service desk agent opens an Incident form where the Category is set to "Hardware". The UI Policy dynamically triggers to make the "Asset Tag" and "Hardware Model" fields mandatory and visible, ensuring technicians capture essential hardware tracking information before saving the record.

Scenario 2: A non-admin user attempts to modify the "State" field of a closed incident. A Client Script validates the current state on submission or field change, instantly throwing an alert and preventing unauthorized modifications to historical ticket data.

Scenario 3: When the Incident Priority is automatically calculated or manually set to "Critical (1)", a UI Policy highlights the incident summary banner in red and mandates the "Business Impact" field, ensuring high-priority outages receive immediate attention and proper impact documentation.

Technical Architecture:

The solution operates on a robust three-tier enterprise service architecture. The Presentation Layer consists of the ServiceNow Service Portal and Native UI rendering dynamic forms. The Business Logic Tier utilizes ServiceNow UI Policies for declarative field-level control and Client Scripts (onChange, onLoad, onSubmit) executed within the browser via the Rhino JavaScript engine. The Data Tier resides in the ServiceNow CMDB and Task/Incident database tables managed by the underlying relational database management system.

Technology	Justification
ServiceNow ITSM	Provides the core ITIL-compliant incident management framework and pre-built data schemas required for rapid deployment.
JavaScript (Client-side)	Enables powerful, asynchronous client-side form validation and dynamic user interactions without page reloads.
ServiceNow UI Policies	Offers a zero-code/low-code configuration engine to manage field visibility, read-only attributes, and mandatory rules efficiently.
GlideForm (g_form) API	Supplies standardized client-side script methods to inspect and manipulate form fields and values seamlessly.

Pre-requisites:

- **ServiceNow Developer Instance:** Active Utah, Vancouver, or Washington DC release instance with admin privileges.
- **JavaScript Fundamentals:** Understanding of client-side scripting concepts, conditional logic, and asynchronous functions.
- **ITIL Foundation Knowledge:** Familiarity with IT service management incident lifecycles and priority matrices.
- **ServiceNow Administration Basics:** Knowledge of form designer, dictionary overrides, and Application Scope.

Project Workflow:

Milestone 1: Environment Setup & Analysis

• Activity 1.1: Provision and configure a personal ServiceNow developer instance.
• Activity 1.2: Analyze standard Incident form layout and identify fields requiring dynamic behavior.

Milestone 2: UI Policy Implementation

• Activity 2.1: Define conditions for field visibility and mandatory requirements based on Category and Priority.
• Activity 2.2: Test and validate UI Policy execution on both Service Portal and Native UI views.

Milestone 3: Client Script Development

• Activity 3.1: Write `onChange` and `onSubmit` client scripts utilizing the `g_form` API for advanced validation.
• Activity 3.2: Implement error handling and user alert prompts to prevent invalid submissions.

Milestone 4: Testing & Deployment

• Activity 4.1: Perform end-to-end user acceptance testing across various user roles and device interfaces.
• Activity 4.2: Package update sets and migrate customizations to the target environment.